NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

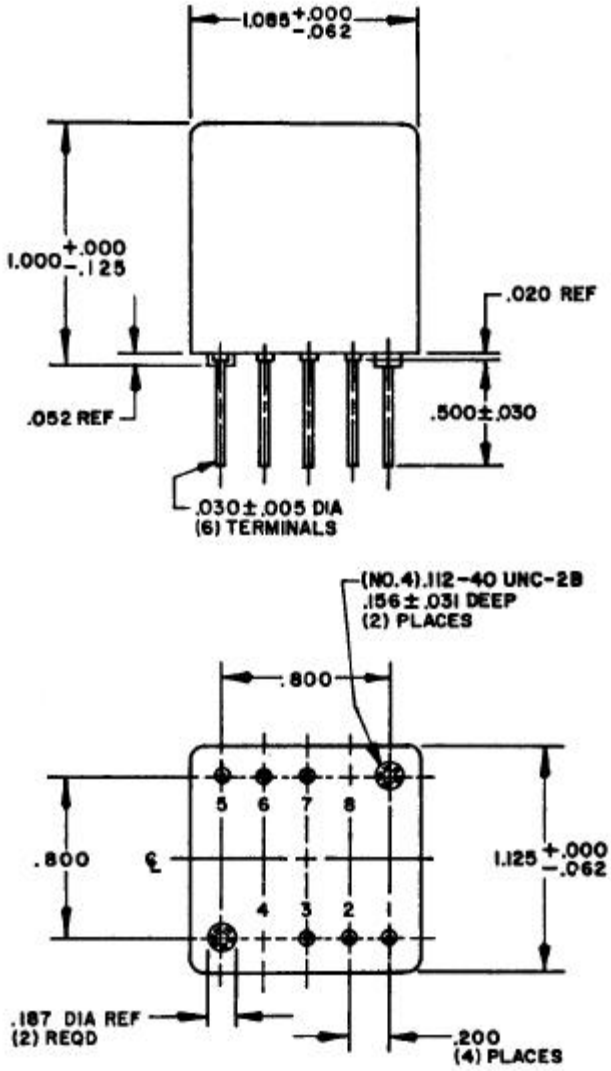
MIL-PRF-27/67A
18 May 1977
SUPERSEDING
MIL-T-27/67
14 April 1976

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY TF5S21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

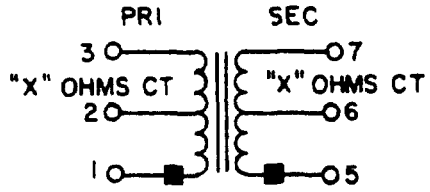
The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.



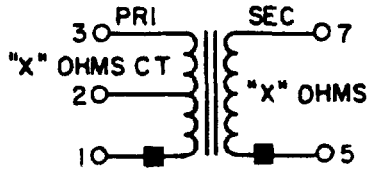
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INCHES	MM	INCHES	MM
005	13	156	3 96
020	.51	187	4 75
.030	.76	200	5 08
031	.79	500	12.70
052	1 32	800	20 32
062	1 57	1.000	25 40
112	2.84	1 085	27 56
125	3.18	1 125	28.58



A



B

WORKING VOLTAGE: "X" V
ALTITUDE . 70 kFT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAMS AND MARKING

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3 Unless otherwise specified, tolerance is ± 010 (25 mm)
4 Marking shall be on sides and/or top of case
5 Electrical values shall be marked as specified in table I, as applicable

FIGURE 1 Dimensions and configuration - Continued

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REQUIREMENTS (When numbers in parentheses, i.e., (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings See table I
Frequency range 60 hertz to 10 kilohertz.
Power level 500 milliwatts maximum at 60 hertz

TABLE I. Electrical ratings 1/

Dash No.	Circuit diagram (see page 4)	Primary impedance ohms (1-3)	Secondary impedance ohms (5-7)	Primary DCR ohms ±25% (1-3)	Secondary DCR ohms ±25% (5-7)	Primary unbalanced DC current max (mA)	Working voltage
01	A	10,000 CT	10,000 CT	700	900	1.0	355
02	A	10,000 CT	600 CT	700	50	1.0	355
03	B	10,000 CT	4	700	0.37	1.0	355
04	B	1,500 CT	15	93	1.2	2.5	175
05	A	1,000 CT	600 CT	65	50	3.0	175
06	B	1,000 CT	4	65	0.37	3.0	175
07	A	600 CT	600 CT	37	50	4.0	175
08	A	600 CT	60 CT	37	5	4.0	175
09	B	600 CT	15	37	1.2	4.0	175
10	B	600 CT	4	37	0.37	4.0	175
11	A	250 CT	600 CT	16	50	7.0	175

1/ Qualification testing and approval to M27/67-01 shall be sufficient to grant qualification approval to M27/67-02 through M27/67-11.

Design and construction
Dimensions and configuration See figure 1
Duty cycle Continuous
Case: Encapsulated
Material Plastic
Terminals Pin (printed circuit)
Material Nickel iron alloy, tin plated.
Diameter 0.03 ±0.005 inch
Length 0.500 ±0.030 inch
Weight 73.71 grams
Altitude 70,000 feet maximum
Operating temperature range -55°C to +130°C

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage
At sea level: 1,000 volts rms (except 500 volts rms on M27/67-04 through M27/67-11)
At barometric pressure 444 volts rms (except 300 volts rms on M27/67-04 through M27/67-11)

Electrical characteristics See table II
Frequency response
Reference frequency 1 kilohertz
Polarity Additive, with terminals 3 and 5 connected

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TABLE II. Electrical characteristics.

Dash No.	Frequency response at 60 Hz to 10 kHz ±1dB			
	Z_S (1-3) ohms	Z_L (5-7) ohms	E_L volts	Primary DC current mA (1-3)
01	10 k	10 k	16.0	0 5
02	10 k	600	16.0	0.5
03	10 k	4	16.0	0 5
04	1.5 k	15	6.0	1.25
05	1.0 k	600	5.0	1.5
06	1.0 k	4	5.0	1.5
07	600	600	4.0	2.0
08	600	60	4.0	2.0
09	600	15	4.0	2.0
10	600	4	4.0	2.0
11	250	600	4.0	3.5

Marking location See figure 1.

Part number M27/67-(dash number from table I).

NOTE Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - ML, WC, ME
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity
Army - EL

Agent
DLA - ES

(Project 5950-0543-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-27/67A TRANSFORMERS, AUDIO FREQUENCY, TF5S21ZZ

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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